

Teaching a neural network to do neutrino physics

NUWRO Workshop 2026

José L. Bonilla

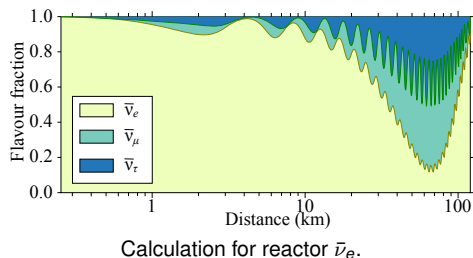
September 18, 2026





- 1 Introduction
- 2 GAN
- 3 Transfer learning
- 4 Summary

There are 3 known “flavours” of neutrinos: ν_e , ν_μ , and ν_τ . The probability of changing from flavour α to flavour β , $p(\alpha \rightarrow \beta) \neq 0$. this probability depends on the initial energy E_ν of the neutrino as well as the neutrino travelled distance.

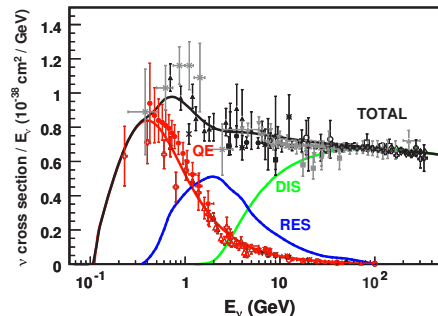


Furthermore, the mechanism that allows the “oscillation” of flavour *eigenstates*, allows for the neutrinos Charge-Parity violation.

Present and future experimental collaborations such as T2K, MicroBooNE, DUNE, and Hyper-Kamiokande goal is to perform precise measurements of the neutrino oscillation parameters.

- Require procedures for E_ν reconstruction from detected final-state particles.
- Rely on Monte Carlo (MC) simulations.
- Limitations due to models assumptions and approximations.

(Thanks to Kajetan for his great talk on the topic on Wednesday!)

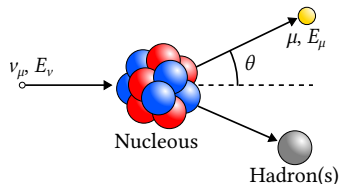


Rev. Mod. Phys. 84 (2012), pp. 1307–1341



- We focus on the development of alternatives that can learn directly from available experimental data.
- We use *conditional Generative Adversarial Networks* (cGANs) to simulate muon neutrino ν_μ scattering and obtain muon kinematic distributions.
- The resultant model can be “extrapolated” to an unseen experimental data set to further train using transfer learning.
- The idea of transfer learning with electron scattering cross sections has been explored before (Graczyk, Kowal, et al: Phys. Rev. Lett. 135, 052502)

In particular, we simulate charged current (CC) muon neutrino-carbon scattering¹. Implemented on Keras 2.15 (Tensorflow) We have created two networks for two cases:



Neutrino-nucleus scattering

- Quasi-elastic (QE) scattering.
- Inclusive (INC) scattering (quasi-elastic and all other process).

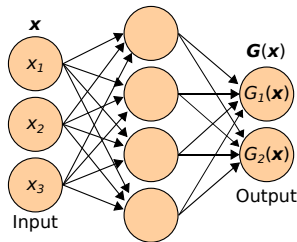
¹ All details: Phys. Rev. D 112, 013007

What are GANs?

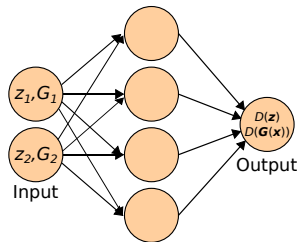


GANs enable training an artificial neural network *Generator* that learns to generate data produced by an unknown or poorly understood mechanism, via a second network *Discriminator* loss.

- A GAN model takes an input latent vector $\mathbf{x}$ of “noise” drawn from a Gaussian distribution and generates samples that match the features of the training data.



Generator



Discriminator

What are GANs?



Real



Discriminator

What are GANs?



Real



Generator



Fake



Discriminator

What are GANs?



Real



Generator



Fake



Discriminator

What are GANs?



Real



Generator

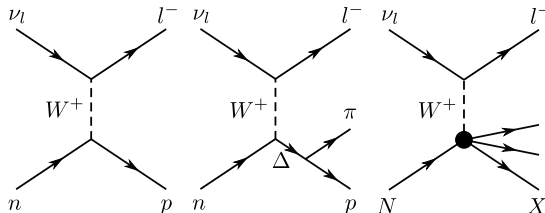


Fake



Discriminator

- The GAN is trained on data generated by the NuWRO 21.09 MC generator to produce scattering events.
- NuWRO has been under development since 2004 at the University of Wrocław.
- It is optimized for the energy range characteristic of accelerator-based neutrino sources (hundreds of MeV to tens of GeV).



Input:

- Neutrino energy E_ν .

Output, we focus on μ kinematics only:

- Muon energy E_μ .
- Muon dispersion angle θ w.r.t neutrino beam.

The longitudinal momentum $p_{\mu,z}$, E_μ , and θ are related as follows:

$$p_{\mu,z} = \sqrt{E_\mu^2 - m_\mu^2} \cos \theta, \quad (1)$$

where m_μ is the muon mass.

We consider neutrino energies from $E_{\nu,min} = 300$ MeV to $E_{\nu,max} = 10$ GeV. For optimization purposes, re-scale the energy variable:

$$E'_\nu = 2 \frac{E_\nu - E_{\nu,min}}{E_{\nu,max} - E_{\nu,min}} - 1,$$

Similarly, for the training of the generator we use a pair (E'_μ, θ') , given by:

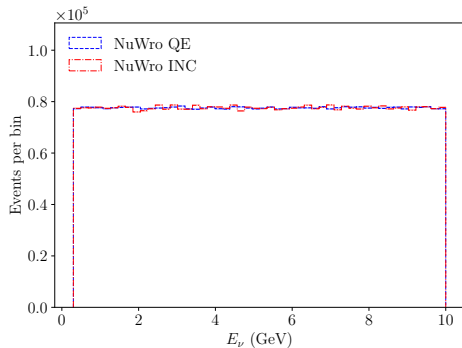
$$E'_\mu = 2 \sqrt{1 - \frac{E_\mu - m_\mu}{\Delta E}} - 1, \quad \theta' = 2 \sqrt{\frac{\theta}{\pi}} - 1,$$

where $\Delta E = E_\nu - 9.8 \text{ MeV} - m_\mu$.

Training sample

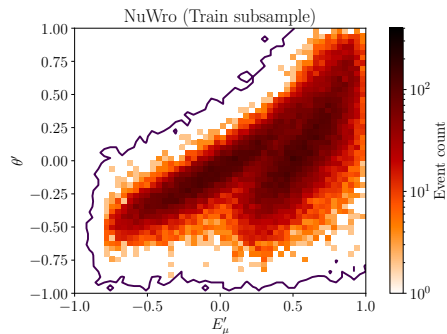


To train the GAN, we sampled $\sim 4\text{M}$ events with NUWRO following a **random uniform** E_ν distribution from 300 MeV to 10 GeV.

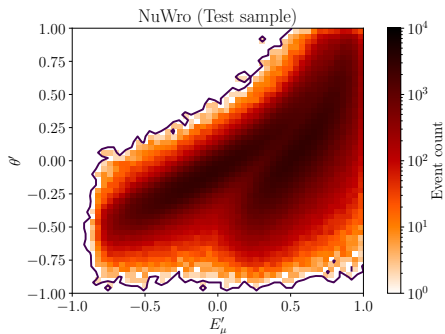


Training samples energy distributions

Even though we can produce a huge amount of data for training, it is very difficult to cover the whole kinematic region with the training dataset.

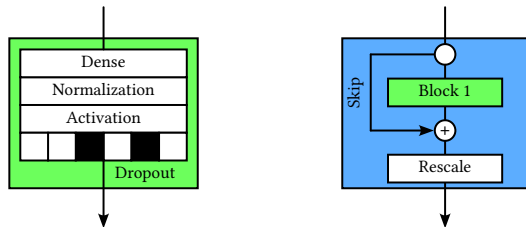


$950 < E_\nu < 1050 \text{ MeV}$



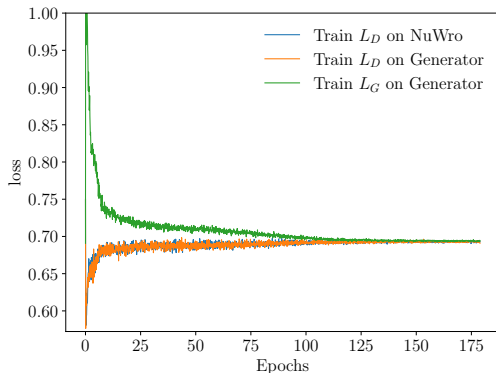
$E_\nu = 1 \text{ GeV}$

The Generator G and Discriminator D are made with basic building blocks as follows:



Which we called `Block 1` and `Block 2`, respectively. Furthermore, we concatenate the initial E'_v information to the input of `Block 1` to create `Block 3`, and then `Block 4`. Additionally, in some of the networks we add Gaussian noise in some blocks.

- We train using the binary cross entropy (bce) loss.
- We monitor the optimization of the generator and discriminator by evaluating their respective loss values.
- The loss curves converge to the same value, corresponding to the configuration where a balance is achieved between G and D.
- The loss values are not enough to evaluate the models quality.



Training loss of INC model

A way of checking the quality of the model is by estimating the *pull*. Given the histograms of events generated by NUWRO and the GAN, the i -th component of the pull is

$$\text{pull}_i = \frac{n_{\text{NuWro},i} - n_{\text{gan},i}}{\sqrt{\sigma_{\text{NuWro},i}^2 + \sigma_{\text{gan},i}^2}}, \quad (2)$$

where $n_{\text{NuWro},i}$ and $n_{\text{gan},i}$ are the bin contents at the i -th bin, and $\sigma_{\text{NuWro},i}^2 = n_{\text{NuWro},i}$ and $\sigma_{\text{gan},i}^2 = n_{\text{gan},i}$ are statistical uncertainties. The bin contents of a histogram are expected to follow the Poisson distribution.

To check the goodness of the model, we evaluate the mean of the absolute value of pulls

$$\text{MAP} = \frac{1}{K} \sum_{i=1}^K |\text{pull}_i| \quad (3)$$

on 2-dimensional (E'_μ, θ') distributions, where K is the number of bins that satisfy $n_{nuw,i} \neq 0$ and $n_{gan,i} \neq 0$.

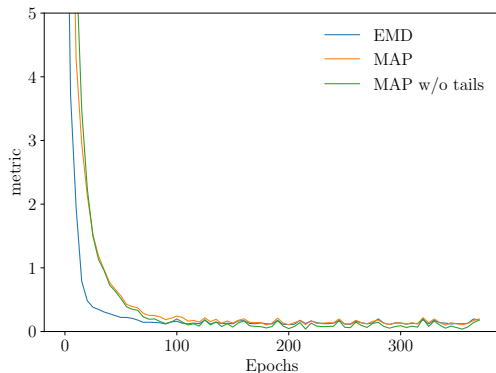
As the G event distribution gets closer to NuWRO's, then Eq. (2) defines random a variable that follows a standard normal distribution $N(0, 1)$ for large $n_{NuWro,i}$ and $n_{gan,i}$. For a random variable $x \sim N(0, 1)$, the expected value of $|x|$ is

$$\mathbb{E}_{x \sim N(0,1)}[|x|] = \sqrt{2/\pi} \approx 0.80. \quad (4)$$

The *Wasserstein distance*, or *Earth Mover's Distance* EMD, is another metric to monitor the quality of the models. It measures how different two given histograms are and how much “work” one must do to redistribute one histogram into another. We evaluate this metric to monitor the quality of the training and final models.



- The MAP and the EMD depend on histogram binning.
- 1M events were generated with both NUWRO and the GAN, with a E'_μ and θ' binning of 50×50 .
- We checked that the MAP between two (normalized) NUWRO samples is ~ 0.8 .
- The EMD computed for two NUWRO samples is ~ 0.03 - 0.04 .



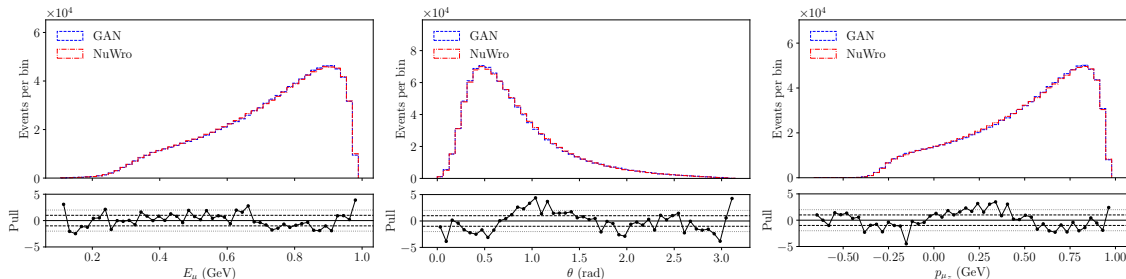
Training performance of QE model



In order to compare the production of events of the GAN, 1M NuWRO events were produced with energies randomly sampled from a uniform distribution in the 300 MeV-10 GeV range, as well as for different values of neutrino energy within the training range,

$E_\nu = 500 \text{ MeV}, 800 \text{ MeV}, 1 \text{ GeV}, 2 \text{ GeV}, 3 \text{ GeV}, 5 \text{ GeV}, 7 \text{ GeV}, 9 \text{ GeV}.$

The G-QE model generates event histograms that closely match those produced by the NuWro. Notably, the distribution of the QE events in muon energy exhibits a single peak.



To test if our GAN model can correctly reproduce the QE and $\Delta(1232)$ resonance peaks, let us now consider event distributions for hadronic invariant mass W

$$W = \sqrt{M_N^2 + 2\omega M_N - Q^2}, \quad (5)$$

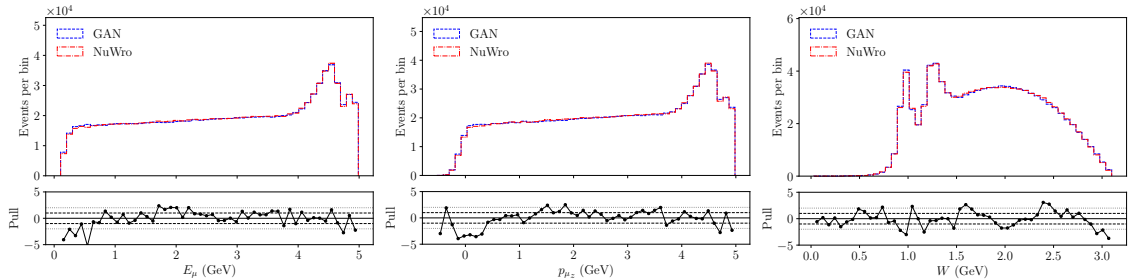
where $\omega = E_\nu - E_\mu$, M_N is the average mass of the nucleon, and Q^2 the four-momentum transfer

$$Q^2 = 2E_\nu(E_\mu - p_{\mu,z}) - m_\mu^2, \quad (6)$$

so the value of W can be obtained from the muon kinematics alone².

²If one disregards the Fermi motion of nucleons in the target nucleus and their binding energies effect.

We observe good agreement between the results from NuWro and G-INC for these variables, both in the quasielastic (QE) region and at the peak of the $\Delta(1232)$ resonance. This consistency extends to the onset of deep inelastic (DIS) region at larger W .



EMD and MAP (with and without tails) computed for the test samples for fixed neutrino energy E_ν and randomly sampled from a uniform E_ν distribution (“All” tag).

E_ν	EMD	MAP	MAP w/o tails
500 MeV	0.15	1.16	1.17
800 MeV	0.14	1.06	1.06
1 GeV	0.14	1.02	1.00
2 GeV	0.14	1.01	0.98
3 GeV	0.13	0.98	0.97
5 GeV	0.13	0.92	0.90
7 GeV	0.15	1.05	1.02
9 GeV	0.11	1.03	0.99
All	0.11	0.90	0.84

QE metrics

E_ν	EMD	MAP	MAP w/o tails
500 MeV	0.16	1.08	1.07
800 MeV	0.16	1.09	1.09
1 GeV	0.15	1.07	1.07
2 GeV	0.15	1.02	1.00
3 GeV	0.14	0.94	0.92
5 GeV	0.14	0.96	0.93
7 GeV	0.16	1.01	0.99
9 GeV	0.15	1.03	1.02
All	0.13	0.89	0.86

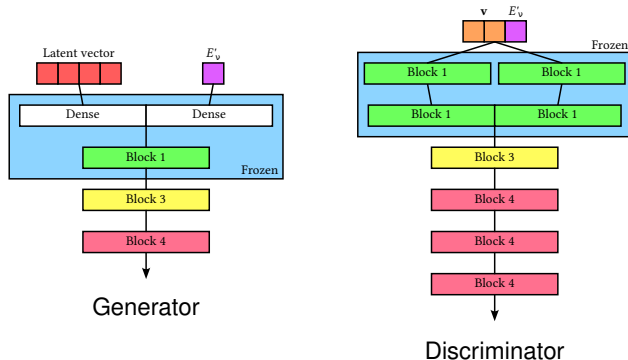
INC metrics



- This technique is useful when data in the target domain is limited, but have a lot of data on a related domain³.

³All details and changes: Phys. Rev. D 113, 053001

- This technique is useful when data in the target domain is limited, but have a lot of data on a related domain³.
- We take advantage of the knowledge obtained from the ν_μ -carbon data, and adjust on a different dataset.
- With TL we “freeze” some layers of the base model and let the rest of the model weights update.



³All details and changes: Phys. Rev. D 113, 053001



- We adapted the ν_μ -carbon model to 3 different scenarios.
 - ▶ neutrino-argon.
 - ▶ antineutrino-carbon.
 - ▶ alternative neutrino-carbon model (Newer NUWRO 25.03).

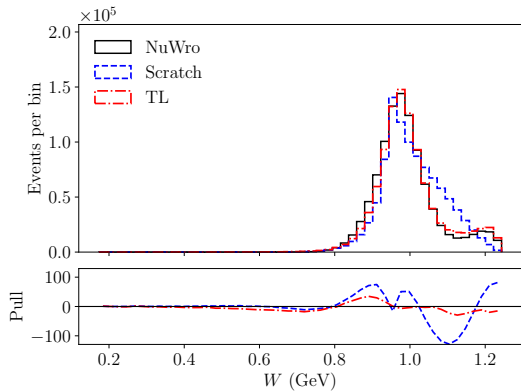


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 - ★ Increased axial masses.
 - ★ QE spectral function was changed to Local Fermi Gas.
 - ★ New Delta pion production model

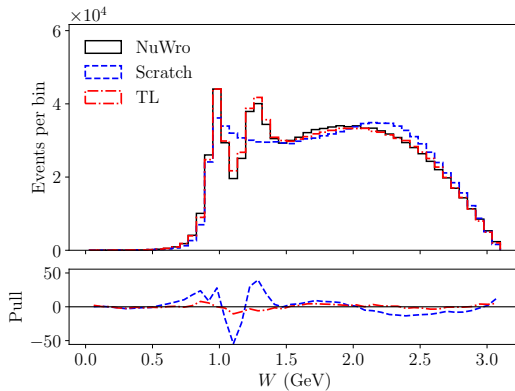


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 - ★ Increased axial masses.
 - ★ QE spectral function was changed to Local Fermi Gas.
 - ★ New Delta pion production model
- This time only INC models were considered, with training samples containing 10k and 100k events.
- We also train the models from scratch using the same datasets to compare performance.

TL results, from ν_μ -carbon to ν_μ -argon

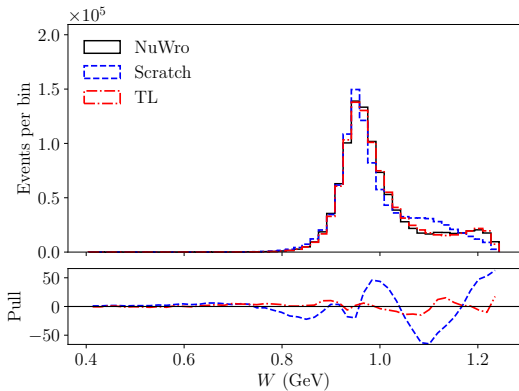


$E_\nu = 500$ MeV

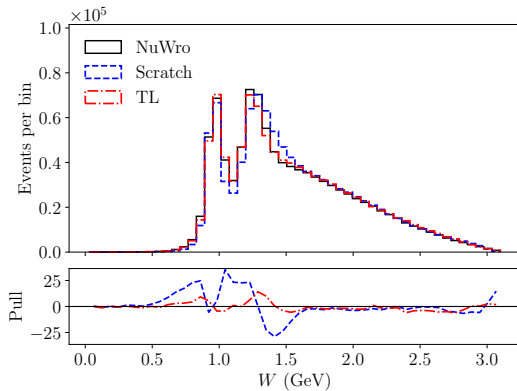


$E_\nu = 5$ GeV

TL results, from ν_μ -carbon to $\bar{\nu}_\mu$ -carbon

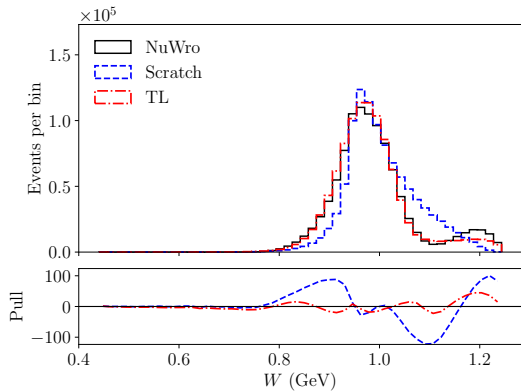


$E_\nu = 500$ MeV

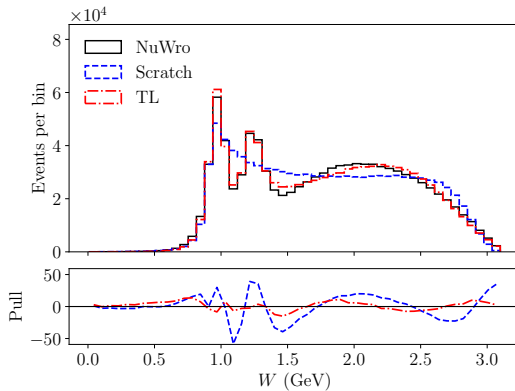


$E_\nu = 5$ GeV

TL results, from ν_μ -carbon to alternative ν_μ -carbon



$E_\nu = 500$ MeV



$E_\nu = 5$ GeV

Can we actually train on data?



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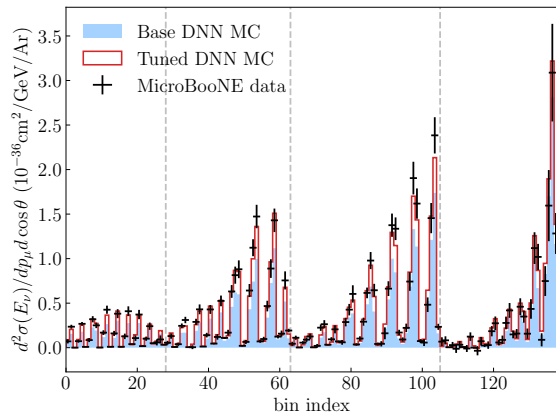


Yes, we can!



Yes, we can!

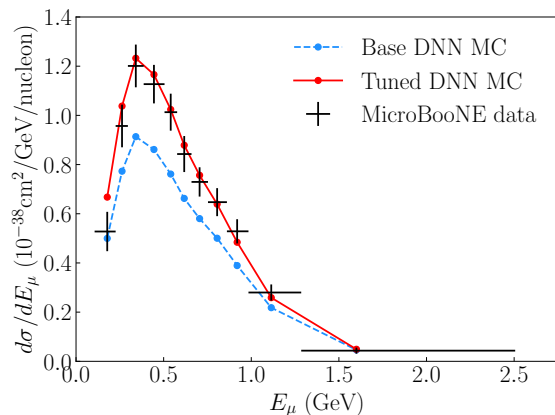
- We need to take into account experimental measurement and statistics.
- Create a new sample taking it into account.
- Proper re-weight.



- We follow a similar procedure as in the training with transfer learning.
- Only the last dense layer of G and D unfrozen during training.
- Improvement respect the base model is evident.
- All details in our upcoming paper!

MicroBooNE inclusive double differential cross section as function of E_ν . Phys. Lett. B 870 (2025) 139939

- We corroborate our results with MicroBooNE results for different differential cross section.



MicroBooNE inclusive differential cross section. Phys.Rev.Lett. 128, 151801

- We have discussed the development of GANs models for simulating quasielastic and inclusive neutrino-nucleus scattering.
- We consider various kinematic distributions of the charged lepton. The models we present successfully reproduce the peak structure in data distributions.
- These models can be adapted to more realistic scenarios after retraining them on experimental data. They can serve as pre-trained models that can be fine-tuned for specific applications.
- We do not need an explicit experimental event by event measurements to train the GAN
- As a side effect, once these models are developed, they generate events significantly faster than “classical” generators. Original inclusive generator model size is 368.7 kB! (and getting smaller)



Demonstration

Thank you!

